

BookMyShow-Scale Ticketing Backend

MySQL schema, sample data, concurrency-safe locking strategy, and P1/P2 SQL

Scope: A theatre has screens, seats, movies and shows. A show has a date/time and a fixed seat inventory. Thousands of users may attempt to reserve the same seats concurrently. Seats are held for a limited period, holds expire automatically, and payment webhooks must be idempotent.

1. Entity model

Entity	Purpose
users	Customer accounts that create bookings.
movies	Movies that can be scheduled.
theatres	Physical theatre/location.
screens	Individual auditorium/screen within a theatre.
seats	Physical seat definition within a screen.
shows	A movie scheduled on a screen at a start time.
show_seats	Per-show seat inventory/state; one row per seat per show.
bookings	Customer reservation/order and its lifecycle.
booking_items	Seats belonging to a booking.
payment_transactions	Payment attempt/provider transaction and idempotency event key.

2. Relationships

The main relationships are: theatre 1:N screens; screen 1:N seats; movie 1:N shows; screen 1:N shows; show 1:N show_seats; seat 1:N show_seats across different shows; user 1:N bookings; booking 1:N booking_items; booking 1:N payment_transactions.

A seat is not globally locked; the seat is locked only for a particular show. The primary key (show_id, seat_id) guarantees exactly one inventory row for a seat in a show.

3. Table definitions

users

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
name	VARCHAR(120)	NOT NULL
email	VARCHAR(255)	NOT NULL, UNIQUE
phone	VARCHAR(30)	NULL
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

movies

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
title	VARCHAR(255)	NOT NULL
duration_minutes	SMALLINT UNSIGNED	NOT NULL
language	VARCHAR(50)	NOT NULL
certificate	VARCHAR(20)	NULL
created_at	DATETIME(6)	NOT NULL DEFAULT

		CURRENT_TIMESTAMP(6)
--	--	----------------------

theatres

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
name	VARCHAR(160)	NOT NULL
address	VARCHAR(500)	NOT NULL
city	VARCHAR(100)	NOT NULL
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

screens

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
theatre_id	BIGINT UNSIGNED	FK theatres.id
name	VARCHAR(80)	NOT NULL
screen_type	VARCHAR(40)	NOT NULL
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

seats

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
screen_id	BIGINT UNSIGNED	FK screens.id
row_label	VARCHAR(10)	NOT NULL
seat_number	SMALLINT UNSIGNED	NOT NULL
seat_type	VARCHAR(30)	NOT NULL
UNIQUE	(screen_id,row_label,seat_number)	one physical seat

shows

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
movie_id	BIGINT UNSIGNED	FK movies.id
screen_id	BIGINT UNSIGNED	FK screens.id
show_date	DATE	NOT NULL
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL
status	ENUM('SCHEDULED','CANCELLED','COMPLETED')	NOT NULL DEFAULT 'SCHEDULED'
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

show_seats

Column	Type	Constraints
show_id	BIGINT UNSIGNED	PK/FK shows.id
seat_id	BIGINT UNSIGNED	PK/FK seats.id
status	ENUM('AVAILABLE','HELD','BOOKED')	NOT NULL DEFAULT 'AVAILABLE'
hold_token	CHAR(36)	NULL
hold_expires_at	DATETIME(6)	NULL
booking_id	BIGINT UNSIGNED	NULL/FK bookings.id

bookings

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
booking_reference	CHAR(16)	NOT NULL, UNIQUE

user_id	BIGINT UNSIGNED	FK users.id
show_id	BIGINT UNSIGNED	FK shows.id
status	ENUM('PENDING','CONFIRMED','EXPIRED','CANCELLED','PAYMENT_FAILED')	NOT NULL
hold_token	CHAR(36)	NOT NULL, UNIQUE
expires_at	DATETIME(6)	NOT NULL
total_amount	DECIMAL(10,2)	NOT NULL
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

booking_items

Column	Type	Constraints
booking_id	BIGINT UNSIGNED	PK/FK bookings.id
seat_id	BIGINT UNSIGNED	PK/FK seats.id
price	DECIMAL(10,2)	NOT NULL

payment_transactions

Column	Type	Constraints
id	BIGINT UNSIGNED	PK
booking_id	BIGINT UNSIGNED	FK bookings.id
provider	VARCHAR(40)	NOT NULL
provider_payment_id	VARCHAR(128)	NULL
provider_event_id	VARCHAR(128)	NOT NULL, UNIQUE
amount	DECIMAL(10,2)	NOT NULL
status	ENUM('RECEIVED','SUCCESS','FAILED','REFUND_REQUIRED','REFUNDED')	NOT NULL
created_at	DATETIME(6)	NOT NULL DEFAULT CURRENT_TIMESTAMP(6)

4. Directly executable MySQL DDL (P1)

```
CREATE DATABASE IF NOT EXISTS bms;
USE bms;
```

```
CREATE TABLE users (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(120) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  phone VARCHAR(30) NULL,
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6)
);
```

```
CREATE TABLE movies (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(255) NOT NULL,
  duration_minutes SMALLINT UNSIGNED NOT NULL,
  language VARCHAR(50) NOT NULL,
  certificate VARCHAR(20) NULL,
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6)
);
```

```
CREATE TABLE theatres (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(160) NOT NULL,
  address VARCHAR(500) NOT NULL,
  city VARCHAR(100) NOT NULL,
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6)
);
```

```
);
```

```
CREATE TABLE screens (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  theatre_id BIGINT UNSIGNED NOT NULL,  
  name VARCHAR(80) NOT NULL,  
  screen_type VARCHAR(40) NOT NULL,  
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),  
  CONSTRAINT fk_screens_theatre FOREIGN KEY (theatre_id) REFERENCES theatres(id),  
  UNIQUE KEY uq_screen_name (theatre_id, name)  
)
```

```
CREATE TABLE seats (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  screen_id BIGINT UNSIGNED NOT NULL,  
  row_label VARCHAR(10) NOT NULL,  
  seat_number SMALLINT UNSIGNED NOT NULL,  
  seat_type VARCHAR(30) NOT NULL,  
  CONSTRAINT fk_seats_screen FOREIGN KEY (screen_id) REFERENCES screens(id),  
  UNIQUE KEY uq_physical_seat (screen_id, row_label, seat_number)  
)
```

```
CREATE TABLE shows (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  movie_id BIGINT UNSIGNED NOT NULL,  
  screen_id BIGINT UNSIGNED NOT NULL,  
  show_date DATE NOT NULL,  
  start_time TIME NOT NULL,  
  end_time TIME NOT NULL,  
  status ENUM('SCHEDULED', 'CANCELLED', 'COMPLETED') NOT NULL DEFAULT 'SCHEDULED',  
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),  
  CONSTRAINT fk_shows_movie FOREIGN KEY (movie_id) REFERENCES movies(id),  
  CONSTRAINT fk_shows_screen FOREIGN KEY (screen_id) REFERENCES screens(id),  
  KEY idx_shows_screen_date_time (screen_id, show_date, start_time),  
  KEY idx_shows_date_screen (show_date, screen_id)  
)
```

```
CREATE TABLE bookings (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  booking_reference CHAR(16) NOT NULL UNIQUE,  
  user_id BIGINT UNSIGNED NOT NULL,  
  show_id BIGINT UNSIGNED NOT NULL,  
  status ENUM('PENDING', 'CONFIRMED', 'EXPIRED', 'CANCELLED', 'PAYMENT_FAILED') NOT NULL DEFAULT 'PENDING',  
  hold_token CHAR(36) NOT NULL UNIQUE,  
  expires_at DATETIME(6) NOT NULL,  
  total_amount DECIMAL(10,2) NOT NULL,  
  created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),  
  CONSTRAINT fk_bookings_user FOREIGN KEY (user_id) REFERENCES users(id),  
  CONSTRAINT fk_bookings_show FOREIGN KEY (show_id) REFERENCES shows(id),  
  KEY idx_bookings_show_status (show_id, status),  
  KEY idx_bookings_expiry (status, expires_at)  
)
```

```
CREATE TABLE show_seats (  
  show_id BIGINT UNSIGNED NOT NULL,  
  seat_id BIGINT UNSIGNED NOT NULL,  
  status ENUM('AVAILABLE', 'HELD', 'BOOKED') NOT NULL DEFAULT 'AVAILABLE',  
  hold_token CHAR(36) NULL,  
  hold_expires_at DATETIME(6) NULL,  
  booking_id BIGINT UNSIGNED NULL,  
  PRIMARY KEY (show_id, seat_id),  
  CONSTRAINT fk_show_seats_show FOREIGN KEY (show_id) REFERENCES shows(id),  
  CONSTRAINT fk_show_seats_seat FOREIGN KEY (seat_id) REFERENCES seats(id),  
  CONSTRAINT fk_show_seats_booking FOREIGN KEY (booking_id) REFERENCES bookings(id),  
  KEY idx_show_seats_release (status, hold_expires_at),  
  KEY idx_show_seats_booking (booking_id)  
)
```

```
CREATE TABLE booking_items (  

```

```

        booking_id BIGINT UNSIGNED NOT NULL,
        seat_id BIGINT UNSIGNED NOT NULL,
        price DECIMAL(10,2) NOT NULL,
        PRIMARY KEY (booking_id, seat_id),
        CONSTRAINT fk_booking_items_booking FOREIGN KEY (booking_id) REFERENCES bookings(id),
        CONSTRAINT fk_booking_items_seat FOREIGN KEY (seat_id) REFERENCES seats(id)
    );

CREATE TABLE payment_transactions (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    booking_id BIGINT UNSIGNED NOT NULL,
    provider VARCHAR(40) NOT NULL,
    provider_payment_id VARCHAR(128) NULL,
    provider_event_id VARCHAR(128) NOT NULL UNIQUE,
    amount DECIMAL(10,2) NOT NULL,
    status ENUM('RECEIVED','SUCCESS','FAILED','REFUND_REQUIRED','REFUNDED') NOT NULL,
    created_at DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
    CONSTRAINT fk_payment_booking FOREIGN KEY (booking_id) REFERENCES bookings(id),
    KEY idx_payment_booking (booking_id),
    UNIQUE KEY uq_provider_payment (provider, provider_payment_id)
);

```

5. Sample entries

```

INSERT INTO users (id,name,email,phone) VALUES
(1,'Aarav Sharma','aarav@example.com','+919900000001'),
(2,'Meera Joshi','meera@example.com','+919900000002');

INSERT INTO movies (id,title,duration_minutes,language,certificate) VALUES
(1,'Example Movie 1',145,'Hindi','UA'),
(2,'Example Movie 2',132,'English','U');

INSERT INTO theatres (id,name,address,city) VALUES
(1,'BMS Phoenix Theatre','123 High Street','Pune');

INSERT INTO screens (id,theatre_id,name,screen_type) VALUES
(1,1,'Screen 1','IMAX'),
(2,1,'Screen 2','2D');

INSERT INTO seats (id,screen_id,row_label,seat_number,seat_type) VALUES
(1,1,'A',1,'REGULAR'),
(2,1,'A',2,'REGULAR'),
(3,1,'A',3,'PREMIUM'),
(4,1,'B',1,'REGULAR'),
(5,1,'B',2,'REGULAR'),
(6,2,'A',1,'REGULAR'),
(7,2,'A',2,'REGULAR');

INSERT INTO shows
(id,movie_id,screen_id,show_date,start_time,end_time,status) VALUES
(101,1,1,'2026-09-22','10:00:00','12:25:00','SCHEDULED'),
(102,1,1,'2026-09-22','14:00:00','16:25:00','SCHEDULED'),
(103,2,2,'2026-09-22','18:30:00','20:42:00','SCHEDULED'),
(104,2,2,'2026-09-23','11:00:00','13:12:00','SCHEDULED');

INSERT INTO show_seats (show_id,seat_id,status) VALUES
(101,1,'AVAILABLE'),(101,2,'AVAILABLE'),(101,3,'AVAILABLE'),
(101,4,'AVAILABLE'),(101,5,'AVAILABLE'),
(102,1,'AVAILABLE'),(102,2,'AVAILABLE'),(102,3,'AVAILABLE'),
(103,6,'AVAILABLE'),(103,7,'AVAILABLE');

```

6. P2 — List all shows for a theatre on a date

```
SELECT
    s.id AS show_id,
    m.title AS movie_title,
    sc.name AS screen_name,
    s.show_date,
    s.start_time,
    s.end_time,
    s.status
FROM shows AS s
JOIN screens AS sc
    ON sc.id = s.screen_id
JOIN movies AS m
    ON m.id = s.movie_id
WHERE sc.theatre_id = :theatre_id
    AND s.show_date = :show_date
    AND s.status = 'SCHEDULED'
ORDER BY s.start_time, sc.name;
```

Example:

```
-- :theatre_id = 1
-- :show_date = '2026-09-22'
```